

PDF ENGINE EDGE-CASE DIAGNOSTIC REPORT

This programmatic test document checks renderer handling for extreme font scaling, uncommon Unicode glyphs, zero-width spaces, and intentional blank page pagination boundaries.

1. MICRO-TYPOGRAPHY & SUB-PIXEL RENDER TEST

Standard Baseline (8pt): The quick brown fox jumps over the lazy dog. 0123456789
Compact Caption (6pt): Legal fine print scaling verification line. Sub-pixel anti-aliasing check.
Micro Text (4pt): Extreme compression test. Verify character spacing, kerning bounds, and glyph overlap.

2. UNICODE GLYPHS & MATHEMATICAL SYMBOLS

Σ ∫ ∞ ∇ ∂ ∞ ∞ ∞
Math & Calculus: Integral, Nabla, Partial Derivatives & Infinity.

♔ ♚ ♜ ♛ ♞ ♟ ♠ ♡ ♢ ♣ ♤ ♥ ♦ ♧ ♨
Chess Symbols: White & Black game piece characters.

⌘ ↵ ⌂ ↩ ↪ ↻ ⚙
UI & System Symbols: Command, Option, Return & Atomic icons.

こんにちは・世界・
CJK & Script Glyphs: Multilingual Unicode block test.

3. FALLBACK FONTS & DENSE CODE STRINGS

```
// Programmatic UTF-8 Boundary & Escape Test const test_matrix = { null_byte: "\0", zero_width: "ABC D", // Zero-width spaces diacritics: "ZaŁgŁ-Ŧext", hex_stream: "0x00 0xFF 0x7F 0x80" };
```

4. EXACT STRUCTURAL BOUNDARY BOX

Margin Boundary Test: This box sits directly above the lower capacity limit. Page 2 is configured as an intentional blank page test for programmatic document pipelines.



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Used for odd-page pagination alignment & printing duplex tests

DIAGNOSTIC TEST RESULTS

Summary of engine performance metrics for edge cases evaluated across Pages 1 and 2.

Test Item	Target Vector	Engine Status
Micro Text Scaling	Font-size down to 2pt	PASS
Unicode Symbol Mapping	Math, CJK, UI Icons	PASS
Empty Page Insertion	Blank Section Flow	PASS